

Logistic Regression (Binary classification) ←

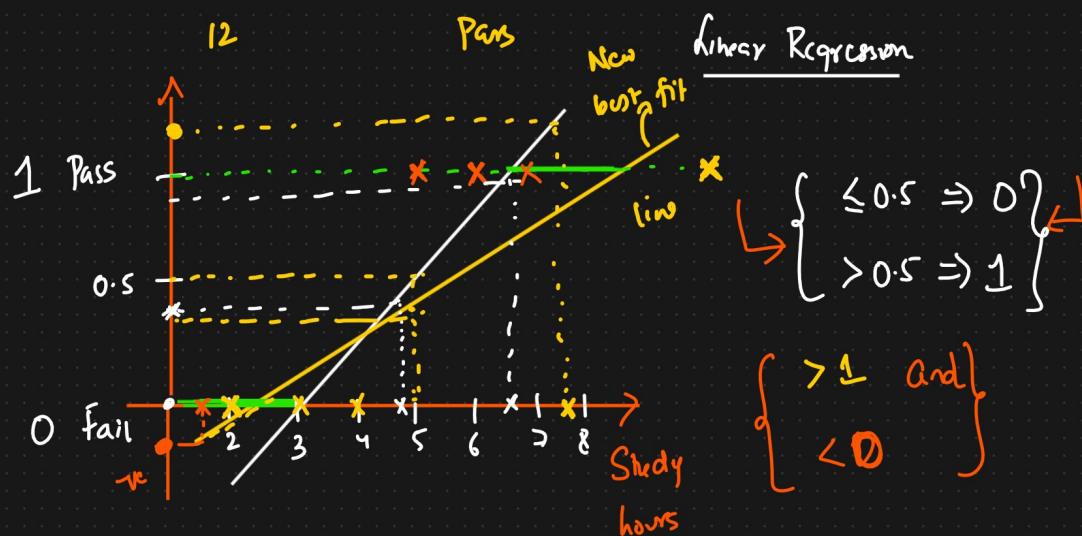
<u>Dataset</u>	<u>Pass/Fail</u>
Study hours	O/p {Binary categories}

2	Fail
3	Fail
4	Fail
5	Pass
6	Pass
7	Pass
12	Pass



Logistic Regression

0 to 1

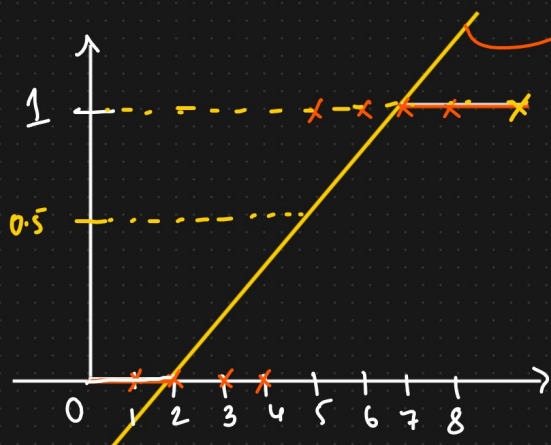


Why we cannot use Linear Regression for Classification?

① Outlier {Best fit line change}

② > 1 and < 0 {Squash line}

How Logistic Regression Solves Classification Problem

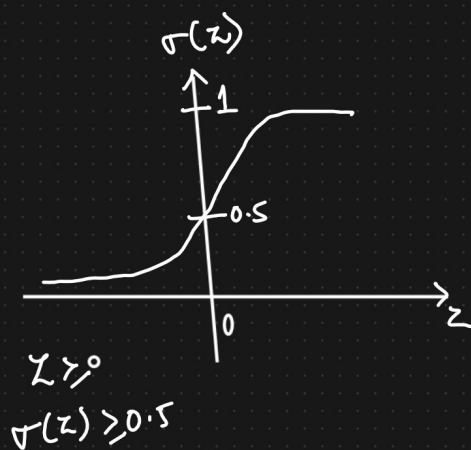


$$h_0(x) = \theta_0 + \theta_1 x_1$$

↓
Best fit line

Sigmoid Activation

$$\frac{1}{1 + e^{-z}}$$



$$h_0(x) = \sigma(\theta_0 + \theta_1 x_1)$$

$$\sigma = \frac{1}{1 + e^{-z}}$$

$$= \sigma(z)$$

$$h_0(x) = \frac{1}{1 + e^{-z}}$$

⇒ Logistic Regression hypothesis

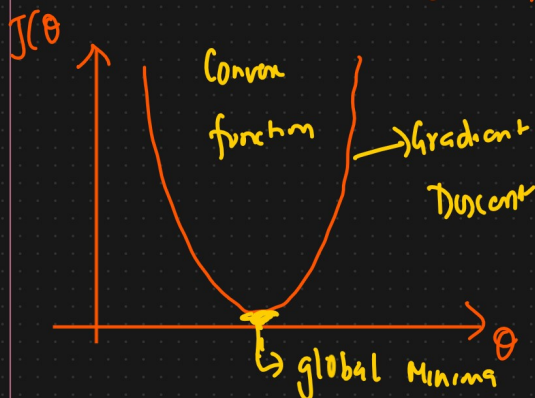
$$z = \theta_0 + \theta_1 x_1$$

Linear Regression Cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$$

$$h_0(x) = \theta_0 + \theta_1 x$$

↓
Convex function



Logistic Regression Cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$$

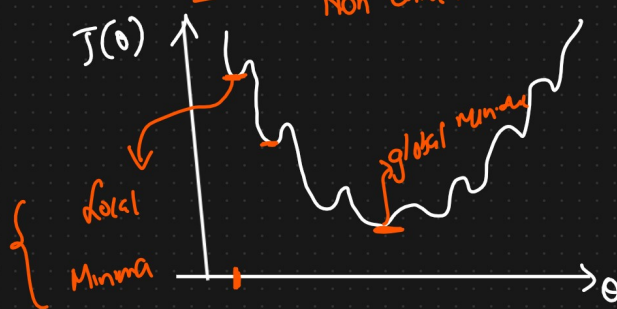
Non Convex function

$$h_0(x) = \frac{1}{1 + e^{-z}}$$

$$z = \theta_0 + \theta_1 x_1$$

Sigmoid

Non Convex



$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \underbrace{(h_{\theta}(x)^{(i)} - y^{(i)})^2}_{\text{cost}(h_{\theta}(x)^{(i)}, y^{(i)})}$$

$$h_{\theta}(x)^{(i)} = \frac{1}{1+e^{-z}} \quad z = \theta_0 + \theta_1 x_i$$

↓

this denote $\text{cost}(h_{\theta}(x)^{(i)}, y^{(i)})$

{ log loss }

$$\text{cost}(h_{\theta}(x)^{(i)}, y^{(i)}) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y=1 \\ -\log(1-h_{\theta}(x)) & \text{if } y=0 \end{cases}$$

↓ convex function

$$\text{cost}(h_{\theta}(x)^{(i)}, y^{(i)}) = -y \log(h_{\theta}(x)) - (1-y) \log(1-h_{\theta}(x))$$

$$J(\theta_0, \theta_1) = -\frac{1}{m} \sum_{i=1}^m (y^{(i)} \log(h_{\theta}(x)^{(i)}) + (1-y^{(i)}) \log(1-h_{\theta}(x)^{(i)}))$$

Minimize cost function $J(\theta_0, \theta_1)$ by changing

θ_0 & θ_1

Convergence Algorithm

Repeat

{

$j=0$ and 1

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

}